

OA-A Variable Harmonization in Detail

This online appendix provides the complete documentation of the bridge implementation summarized in print Appendix A. Section OA-A.1 documents the classification logic of the legacy salaried proxy and the conceptual asymmetry it carried across occupational categories. Section OA-A.2 reports the full inventory of variables added, modified, and deleted in the 2024Q4 redesign. Section OA-A.3 provides the block-by-block interpretation of the new content. Sections OA-A.4 and OA-A.5 present the truth tables for the categorical (64 scenarios) and continuous (250 scenarios) reconstruction algorithms summarized in print Appendix A.2.

OA-A.1 Legacy salaried proxy: classification logic

Before the 2024Q4 redesign, the EPH did not observe labour formality through an administrative registration record covering all workers. The standard historical measure relied on an employee-side proxy: the variable `desc_jubilatorio_asalariado` (DJA), constructed from PP07H, which records whether a salaried worker reports having pension contributions withheld from their wages. A response of “Si” identified a registered (formal) salaried job, “No” an unregistered (informal) one. This proxy is informative for employees because withholding is a feature of the employer–employee relationship, but for non-wage workers the question is conceptually ill-posed: employers and the self-employed do not have contributions withheld from a wage by a third party in the same way. A formal independent worker may contribute directly without wage withholding, in which case a “No” answer to DJA does not imply lack of formal status; an informal independent worker is correctly captured by a “No corresponde” routing flag, but so is a formal one whose contributions are simply not made via payroll.

The legacy code applies three classification rules in sequence to occupied workers (`condicion_actividad == "Ocupado"`). *Formal* is assigned when DJA equals “Si” and `categoria_ocupacional` is one of {Empleado, Patrón, Cuenta Propia}: conceptually correct for employees, mechanically applied to non-wage workers in cases of reporting noise or coding irregularities. *Informal* is assigned when DJA equals “No”: again, correct for employees, ambiguous for non-wage workers. *Non applicable* is assigned when DJA equals “No corresponde”: most employers and the self-employed fall here in a correctly routed file, but the residual category cannot distinguish formal from informal non-wage workers. Home production workers are not classified by the legacy rule; under the redesign they are informal by definition. Table OA-A.1 summarizes the resulting classification across occupational categories.

Table OA-A.1: Legacy classification under `desc_jubilatorio_asalariado`

Occ. category	Actual condition	DJA value		Code output	Conceptually correct?
Employee	Registered with employer withholding	“Si”		Formal	Yes
	Unregistered	“No”		Informal	Yes
Self-employed	Registered with own contributions	“No”/“No resp.”	cor-	Informal/NA	No
	Unregistered	“No”/“No resp.”	cor-	Informal/NA	Ambiguous
Employer	Formal production unit	“No corresp.”		Non applicable	No
	Informal production unit	“No”/“No resp.”	cor-	Informal/NA	Ambiguous
Home production	Informal by definition	“No corresp.”		Non applicable	No

Notes. The legacy code is best understood as a measure of registered versus unregistered *salaried* employment, not as a complete measure of labour informality. Within the employee branch it is a useful and historically consistent proxy. Outside that branch it either misclassifies workers or leaves them in a residual non-applicable category. The asymmetry is the central reason for the bridge: the redesign does not merely relabel the old variable; it expands the observable concept from an employee-side withholding proxy to a broader classification that also covers non-wage work through production-unit characteristics (Appendix A.3).

OA-A.2 Full inventory of new, modified, and deleted variables

This subsection lists the EPH content affected by the 2024Q4 redesign. Three tables organize the inventory: Table OA-A.2 reports variables added or disaggregated, with notes on whether each presents an interpretation issue and whether it is the disaggregated counterpart of a deleted aggregate; Table OA-A.3 reports variables that survived the redesign but had their internal coding modified; Table OA-A.4 reports variables that were dropped from the registry, some of which still appear in the microdata files (a registry–database inconsistency addressed in print Section A.4).

Table OA-A.2: Variables added or disaggregated in the 2024Q4 EPH redesign

Variable	Description	Interpretation issue	Disaggregates a deleted aggregate
Non-labour income sources			
V2_01	Contributory retirement or pension	No	No
V21_01	Bonus from contributory retirement or pension	No	Yes
V22_01	Retroactive payment from contributory retirement or pension	No	Yes
V2_02	Homemaker pension	No	No
V21_02	Bonus from homemaker pension	No	Yes
V22_02	Retroactive payment from homemaker pension	No	Yes
V2_03	Other pensions	No	No
V21_03	Bonus from other pensions	No	Yes
V22_03	Retroactive payment from other pensions	No	Yes
V5_01	AUH subsidy	Yes	Yes
V5_02	Government subsidy	Yes	Yes
V5_03	Charity subsidy	Yes	Yes
V11_01	Government study grant	Yes	Yes
V11_02	Private-sector study grant	Yes	Yes
Activity status and search behavior			
EMPLE0	Formal or informal employment	No	No
SECTOR	Sector of formal or informal employment	No	No
PP02A	Immediate availability to work (hypothetical)	No	No
PP02B	Searched for a job this month	No	No
PP02D	Searches for a job by doing something on their own	No	No
PP02F	Searched for a job this month	Yes	No
PP02G	Immediate availability to work (effective)	Yes	No
PP03K	Reason for job search (employed)	No	No
Main occupation: public-sector and self-employment			
PP04A1	Level of public-sector job	No	No
PP05B3	Issuance of invoices at the independent worker's workplace	No	No
PP05I	Type of social-security contribution by the independent worker	No	No
PP05J	Reason why the independent worker did not contribute	No	No
PP05K	Issuance of invoices at the independent worker's workplace (alt.)	Yes	No
PP06E1	Uses accounting services for independent work	No	No
PP06K	Work dedication in the month (independent worker)	No	No
PP06K_SEM	Workdays per week (independent worker)	No	No
PP06K_MES	Workdays in the month (independent worker)	No	No
PP06L	Work hours per day (independent worker)	No	No
Main occupation: employees			
PP07B1_01	Receives a plan for this job (employees, non-domestic)	No	No
PP07F1_1	Employee must use their own machinery	No	No
PP07F1_2	Employee must use their own premises	No	No
PP07F1_3	Employee must use their own vehicle	No	No
PP07I2	Employer makes at least one pension deduction	No	No
PP07I3	Employer issues invoices	No	No
PP07I4	Employer has an accountant	No	No
PP07L	Employee's payslip covers the full amount or only a part	No	No
PP07M	Share of the employee's pay not included on payslip	No	No
PP08G	Work dedication in the month (employee)	No	No
PP08G_DSEM	Workdays per week (employee)	No	No
PP08G_DMES	Workdays in the month (employee)	No	No
PP08H	Work hours per day (employee)	No	No
Unemployment			
PP10B1	Reason for unemployment: has just started to look	No	No
PP10B2	Reason for unemployment: age	No	No
PP10B3	Reason for unemployment: lack of work in specialty	No	No
PP10B4	Reason for unemployment: no experience/training	No	No
PP10B5	Reason for unemployment: lacks connections	No	No
PP10B6	Reason for unemployment: no jobs available	No	No
PP10B7	Reason for unemployment: cannot afford to job-hunt	No	No
PP10B8	Reason for unemployment: suspended	No	No
PP10B9	Reason for unemployment: other	No	No
PP10B10	Reason for unemployment: does not know	No	No
PP11L2	Stability of last job (yes/no/DK)	Yes	No
Non-labor income amounts			
V2_01_M	Income from contributory retirement or pension	No	Yes
V2_02_M	Income from homemaker pension	No	Yes
V2_03_M	Income from other pensions	No	Yes
V5_01_M	Income from AUH subsidy	No	Yes
V5_02_M	Income from government subsidy	No	Yes
V5_03_M	Income from charity subsidy	No	Yes
V11_01_M	Income from government study grant	Yes	Yes
V11_02_M	Income from private-sector study grant	Yes	Yes
V21_01_M	Income from bonus of contributory retirement or pension	No	Yes
V22_01_M	Income from retroactive payment of contributory retirement or pension	No	Yes
V21_02_M	Income from bonus of homemaker pension	No	Yes
V22_02_M	Income from retroactive payment of homemaker pension	No	Yes
V21_03_M	Income from bonus of other pensions	No	Yes
V22_03_M	Income from retroactive payment of other pensions	No	Yes
Per-capita household income deciles			
P_DECCF	Decile of per-capita household income, TOTAL	No	No
P_IDECCF	Decile of per-capita household income, INTERIOR	No	No
P_RDECCF	Decile of per-capita household income, REGION	No	No
P_GDECCF	Decile of per-capita household income, LARGE AGGLOM.	No	No
P_PDECCF	Decile of per-capita household income, SMALL AGGLOM.	No	No
P_ADECCF	Decile of per-capita household income, AGGLOMERATED	No	No

Table OA-A.3: Variables retained but with modified internal coding

Variable	Description and modification
PP04G	Physical place of main occupation; previous code 1 (shop/office/establishment/business/workshop/farm) disaggregated into codes 11 (shop/office/establishment/business/workshop), 12 (this dwelling, exclusive space), and 13 (farm)
PP07E	Type of work as employee; previous code 1 (employment plan) deleted; remaining codes preserved (probationary period; scholarship/internship/apprenticeship; none of these). New variable PP07B1_01 captures concurrent receipt of social-programme payments
PP07F5	Employee receives no labor benefit; “Yes” explicitly coded 1 in the manual (previously 5)
PP11B1	Domestic-service classification; “Other” (code 2) removed from manual but appears in microdata
PP11L1	Stability of the job left (unemployed branch); previous three-level question (gig vs. permanent vs. DK) reassigned to PP11L2; PP11L1 now a yes/no question with new content

Table OA-A.4: Variables deleted from the EPH as of 2024Q4

Variable	Description
<i>Non-labour income sources (Y/N)</i>	
V11	Study grant
V21	Bonus from retirement or pension
V22	Retroactive payment from retirement or pension
V5	Cash subsidy or social assistance
<i>Non-labor income amounts</i>	
V11_M	Income from study grant
V21_M	Income from retirement or pension
V21_M	Income from bonus of retirement or pension
V5_M	Income from cash subsidy or social assistance
T_VI	Total non-labor income (listed as deleted in registry; persists in microdata — see Section A.4)

OA-A.3 Block-by-block interpretation of new content

The variables added in the 2024Q4 redesign cluster into thematic blocks that together expand the observable concept of labour formality and broaden the diagnostic content of the survey. The following description summarizes what each block measures, what it adds relative to the legacy instrument, and what aspect of the bridge it informs.

Non-labour income sources and amounts. Variables V2_01–V2_03, V21_01–V21_03, V22_01–V22_03, V5_01–V5_03, and V11_01–V11_02 disaggregate transfer types (retirement, pension, social plans, scholarships) and their components (bonuses, retroactives) previously grouped in V2, V21, V22, V5, and V11. Amount counterparts on the person file mirror this fan-out (e.g., V11_01_M, V21_M → component-level _M variables). The block enables finer analysis of household reliance on transfers and complements the harmonization rule of Section OA-A.5: monetary aggregates removed from the redesign are reconstructed from components subject to the documented control-variable consistency.

Activity status and search behavior. Variables EMPLE0, SECTOR, PP02A–PP02G expand the measurement of activity status, hypothetical and effective availability for work, and active job search. EMPLE0 and SECTOR are derived fields constructed by INDEC from the underlying primitives; the bridge does not use them directly because they retain DK/NR categories with non-trivial mass. Instead, the dual-path classification of print Section A.3 reconstructs both fields from the primitives.

Reasons for job search among the employed. PP03K captures motivations for on-the-job search (switching, income supplementation, instability), supporting analysis of voluntary and involuntary mobility margins.

Public-sector employment. PP04A1 disaggregates public-sector workers by level of government (national, provincial, municipal). The block does not enter the formality classification but supports labour-market heterogeneity analyses.

Self-employment formalization. PP05B3, PP05I, PP05J, PP05K measure unit-level formalization signals for the self-employed: invoicing capacity, type of social-security contribution, and reasons for not contributing. This block is the central source of the dual-path concept expansion (Type 3 in Section A.1) and is the substantive content that the legacy proxy could not observe.

Time and accounting use among the self-employed. PP06E1, PP06K, PP06K_SEM, PP06K_MES, PP06L measure work intensity, the use of accounting services, and the temporal structure of self-employment. PP06E1 is the third indicator in the dual-path rule for non-wage workers; the time-use items support analyses of effort intensity and hourly earnings.

Employee-side diagnostic items. For employees in non-domestic private-sector jobs, PP07B1_01 flags concurrent social-programme receipt; PP07F1_1–PP07F1_3 record whether the worker must provide own tools, premises, or vehicle; PP07I2–PP07I4 verify employer-side formality conditions (pension deductions, invoicing, accounting); PP07L–PP07M record payslip coverage and the off-payslip share of compensation. These items are diagnostic only and do not enter the binary formality classification, which continues to rest on PP07H alone (print Section A.3).

Time use among employees. PP08G, PP08G_DSEM, PP08G_DMES, PP08H refine measurement of days and hours worked by employees, supporting underemployment analysis.

Reasons for unemployment. PP10B1–PP10B10 expand the catalogue of reasons for unemployment to ten categories (age, lack of experience, lack of connections, suspension, search costs, others), supporting analyses that distinguish frictional from structural unemployment.

Stability of last job. PP11L2 captures the stability of the last job among the unemployed (permanent vs. temporary), redesignated from the legacy PP11L1; the role swap is the source of the registry–database inconsistency documented in print Section A.4.

Per-capita household income deciles. P_DECCF, P_IDECCF, P_RDECCF, P_GDECCF, P_PDECCF, P_ADECCF assign each individual the per-capita family income decile of their household across alternative geographic universes (national total, interior, region, large/small agglomerations, agglomerated areas), facilitating individual-level inequality analysis.

OA-A.4 Categorical reconstruction: 64-scenario truth table

Section A.2 of the print Appendix A formalizes the hierarchical rule for reconstructing legacy categorical aggregates from three disaggregated components, each taking values in $\{0, 1, 2, 9\}$. The 64 possible scenarios that result from the four-value space and three components are enumerated in Table OA-A.5. The rule satisfies three properties: (i) the affirmation rule dominates (any component = 1 assigns $E = 1$); (ii) under no affirmation, the combination of zeros and nines is resolved by the rule that two or more 9s assign $E = 9$, otherwise $E = 0$; (iii) the residual scenario assigns $E = 2$.

Table OA-A.5: Assignment scenarios for reconstructing a three-component categorical variable

Scenario	Variables (N)			Assigned E	Scenario	Variables (N)			Assigned E
	Var 1	Var 2	Var 3			Var 1	Var 2	Var 3	
1	0	0	0	0	33	0	0	2	2
2	1	0	0	1	34	1	0	2	1
3	2	0	0	2	35	2	0	2	2
4	9	0	0	0	36	9	0	2	2
5	0	1	0	1	37	0	1	2	1
6	1	1	0	1	38	1	1	2	1
7	2	1	0	1	39	2	1	2	1
8	9	1	0	1	40	9	1	2	1
9	0	2	0	2	41	0	2	2	2
10	1	2	0	1	42	1	2	2	1
11	2	2	0	2	43	2	2	2	2
12	9	2	0	2	44	9	2	2	2
13	0	9	0	0	45	0	9	2	2
14	1	9	0	1	46	1	9	2	1
15	2	9	0	2	47	2	9	2	2
16	9	9	0	9	48	9	9	2	2
17	0	0	1	1	49	0	0	9	0
18	1	0	1	1	50	1	0	9	1
19	2	0	1	1	51	2	0	9	2
20	9	0	1	1	52	9	0	9	9
21	0	1	1	1	53	0	1	9	1
22	1	1	1	1	54	1	1	9	1
23	2	1	1	1	55	2	1	9	1
24	9	1	1	1	56	9	1	9	1
25	0	2	1	1	57	0	2	9	2
26	1	2	1	1	58	1	2	9	1
27	2	2	1	1	59	2	2	9	2
28	9	2	1	1	60	9	2	9	2
29	0	9	1	1	61	0	9	9	9
30	1	9	1	1	62	1	9	9	1
31	2	9	1	1	63	2	9	9	2
32	9	9	1	1	64	9	9	9	9

Notes. Coding: 1 = Yes, 2 = No, 0 = Not applicable, 9 = DK/NR. Scenarios are enumerated as the Cartesian product of the four values across the three components. The column E reports the assignment to the reconstructed legacy aggregate under the rule of Section A.2: any 1 dominates; in the absence of any 1, two or more 9s assign 9 and otherwise the result is 0; the residual is 2.

OA-A.5 Continuous reconstruction: 250-scenario truth table

For continuous income variables, each component takes positive values or one of three negative codes (-7 did not have that occupation, -8 no income in reference month, -9 DK/NR). The reconstruction rule applies absolute priority to positive income, cross-validates against the corresponding categorical control flag (which takes the values “No” or “Other”), and otherwise propagates the most informative negative code by the hierarchy $-9 \succ -8 \succ -7$. Table OA-A.6 enumerates the 250 valid scenarios that arise under the four positive/negative-code values across three components and the two control-variable values; in the table the positive case is represented by the scalar 100, which stands for any strictly positive amount.

Table OA-A.6: Assignment scenarios for reconstructing a three-component continuous non-labour-income aggregate

Scen.	Variables (N)			Control (cat.)	Assigned E	Scen.	Variables (N)			Control (cat.)	Assigned E	Scen.	Variables (N)			Control (cat.)	Assigned E
	V1	V2	V3				V1	V2	V3				V1	V2	V3		
1	100	100	100	No	300	85	-9	0	-8	No	-7	168	-7	-8	0	Other	-8
2	0	100	100	No	200	86	100	-7	-8	No	100	169	-8	-8	0	Other	-8
3	-7	100	100	No	200	87	0	-7	-8	No	-7	170	-9	-8	0	Other	-9
4	-8	100	100	No	200	88	-7	-7	-8	No	-7	171	100	-9	0	Other	100
5	-9	100	100	No	200	89	-8	-7	-8	No	-7	172	0	-9	0	Other	-9
6	100	0	100	No	200	90	-9	-7	-8	No	-7	173	-7	-9	0	Other	-9
7	0	0	100	No	100	91	100	-8	-8	No	100	174	-8	-9	0	Other	-9
8	-7	0	100	No	100	92	0	-8	-8	No	-7	175	-9	-9	0	Other	-9
9	-8	0	100	No	100	93	-7	-8	-8	No	-7	176	100	100	-7	Other	200
10	-9	0	100	No	100	94	-8	-8	-8	No	-7	177	0	100	-7	Other	100
11	100	-7	100	No	200	95	-9	-8	-8	No	-7	178	-7	100	-7	Other	100
12	0	-7	100	No	100	96	100	-9	-8	No	100	179	-8	100	-7	Other	100
13	-7	-7	100	No	100	97	0	-9	-8	No	-7	180	-9	100	-7	Other	100
14	-8	-7	100	No	100	98	-7	-9	-8	No	-7	181	100	0	-7	Other	100
15	-9	-7	100	No	100	99	-8	-9	-8	No	-7	182	0	0	-7	Other	0
16	100	-8	100	No	200	100	-9	-9	-8	No	-7	183	-7	0	-7	Other	0
17	0	-8	100	No	100	101	100	100	-9	No	200	184	-8	0	-7	Other	-8
18	-7	-8	100	No	100	102	0	100	-9	No	100	185	-9	0	-7	Other	-9
19	-8	-8	100	No	100	103	-7	100	-9	No	100	186	100	-7	-7	Other	100
20	-9	-8	100	No	100	104	-8	100	-9	No	100	187	0	-7	-7	Other	0
21	100	-9	100	No	200	105	-9	100	-9	No	100	188	-7	-7	-7	Other	-7
22	0	-9	100	No	100	106	100	0	-9	No	100	189	-8	-7	-7	Other	-8
23	-7	-9	100	No	100	107	0	0	-9	No	-7	190	-9	-7	-7	Other	-9
24	-8	-9	100	No	100	108	-7	0	-9	No	-7	191	100	-8	-7	Other	100
25	-9	-9	100	No	100	109	-8	0	-9	No	-7	192	0	-8	-7	Other	-8
26	100	100	0	No	200	110	-9	0	-9	No	-7	193	-7	-8	-7	Other	-8
27	0	100	0	No	100	111	100	-7	-9	No	100	194	-8	-8	-7	Other	-8
28	-7	100	0	No	100	112	0	-7	-9	No	-7	195	-9	-8	-7	Other	-9
29	-8	100	0	No	100	113	-7	-7	-9	No	-7	196	100	-9	-7	Other	100
30	-9	100	0	No	100	114	-8	-7	-9	No	-7	197	0	-9	-7	Other	-9
31	100	0	0	No	100	115	0	-7	-9	No	-7	198	-7	-9	-7	Other	-9
32	0	0	0	No	0	116	100	-8	-9	No	100	199	-8	-9	-7	Other	-9
33	-7	0	0	No	-7	117	0	-8	-9	No	-7	200	-9	-9	-7	Other	-9
34	-8	0	0	No	-7	118	-7	-8	-9	No	-7	201	100	100	-8	Other	200
35	-9	0	0	No	-7	119	-8	-8	-9	No	-7	202	0	100	-8	Other	100
36	100	-7	0	No	100	120	-9	-8	-9	No	-7	203	-7	100	-8	Other	100
37	0	-7	0	No	-7	121	100	-9	-9	No	100	204	-8	100	-8	Other	100
38	-7	-7	0	No	-7	122	0	-9	-9	No	-7	205	-9	100	-8	Other	100
39	-8	-7	0	No	-7	123	-7	-9	-9	No	-7	206	100	0	-8	Other	100
40	-9	-7	0	No	-7	124	-8	-9	-9	No	-7	207	0	0	-8	Other	-8
41	100	-8	0	No	100	125	-9	-9	-9	No	-7	208	-7	0	-8	Other	-8
42	0	-8	0	No	-7	126	100	100	100	Other	300	209	-8	0	-8	Other	-8
43	-7	-8	0	No	-7	127	0	100	100	Other	200	210	-9	0	-8	Other	-9
44	-8	-8	0	No	-7	128	-7	100	100	Other	200	211	100	-7	-8	Other	100
45	-9	-8	0	No	-7	129	-8	100	100	Other	200	212	0	-7	-8	Other	-8
46	100	-9	0	No	100	130	-9	100	100	Other	200	213	-7	-7	-8	Other	-8
47	0	-9	0	No	-7	131	100	0	100	Other	200	214	-8	-7	-8	Other	-8
48	-7	-9	0	No	-7	132	0	0	100	Other	100	215	-9	-7	-8	Other	-9
49	-8	-9	0	No	-7	133	-7	0	100	Other	100	216	100	-8	-8	Other	100
50	-9	-9	0	No	-7	134	-8	0	100	Other	100	217	0	-8	-8	Other	-8
51	100	100	-7	No	200	135	-9	0	100	Other	100	218	-7	-8	-8	Other	-8
52	0	100	-7	No	100	136	100	-7	100	Other	200	219	-8	-8	-8	Other	-8
53	-7	100	-7	No	100	137	0	-7	100	Other	100	220	-9	-8	-8	Other	-9
54	-8	100	-7	No	100	138	-7	-7	100	Other	100	221	100	-9	-8	Other	100
55	-9	100	-7	No	100	139	-8	-7	100	Other	100	222	0	-9	-8	Other	-9
56	100	0	-7	No	100	140	-9	-7	100	Other	100	223	-7	-9	-8	Other	-9
57	0	0	-7	No	-7	141	100	-8	100	Other	200	224	-8	-9	-8	Other	-9
58	-7	0	-7	No	-7	142	0	-8	100	Other	100	225	-9	-9	-8	Other	-9
59	-8	0	-7	No	-7	143	-7	-8	100	Other	100	226	100	100	-9	Other	200
60	-9	0	-7	No	-7	144	-8	-8	100	Other	100	227	0	100	-9	Other	100
61	100	-7	-7	No	100	145	-9	-8	100	Other	100	228	-7	100	-9	Other	100
62	0	-7	-7	No	-7	146	100	-9	100	Other	200	229	-8	100	-9	Other	100
63	-7	-7	-7	No	-7	147	0	-9	100	Other	100	230	-9	100	-9	Other	100
64	-8	-7	-7	No	-7	148	-7	-9	100	Other	100	231	100	0	-9	Other	100
65	-9	-7	-7	No	-7	149	-8	-9	100	Other	100	232	0	0	-9	Other	-9
66	100	-8	-7	No	100	150	-9	-9	100	Other	100	233	-7	0	-9	Other	-9
67	0	-8	-7	No	-7	151	100	100	0	Other	200	234	-8	0	-9	Other	-9
68	-7	-8	-7	No	-7	152	0	100	0	Other	100	235	-9	0	-9	Other	-9
69	-8	-8	-7	No	-7	153	-7	100	0	Other	100	236	100	-7	-9	Other	100
70	-9	-8	-7	No	-7	154	-8	100	0	Other	100	237	0	-7	-9	Other	-9
71	100	-9	-7	No	100	155	-9	100	0	Other	100	238	-7	-7	-9	Other	-9
72	0	-9	-7	No	-7	156	100	0	0	Other	100	239	-8	-7	-9	Other	-9
73	-7	-9	-7	No	-7	157	0	0	0	Other	0	240	-9	-7	-9	Other	-9
74	-8	-9	-7	No	-7	158	-7	0	0	Other	0	241	100	-8	-9	Other	100
75	-9	-9	-7	No	-7	159	-8	0	0	Other	-8	242	0	-8	-9	Other	-9
76	100	100	-8	No	200	160	-9	0	0	Other	-9	243	-7	-8	-9	Other	-9
77	0	100	-8	No	100	161	100	-7	0	Other	100	244	-8	-8	-9	Other	-9
78	-7	100	-8	No	100	162	0	-7	0	Other	0	245	-9	-8	-9	Other	-9
79	-8	100	-8	No	100	163	-7	-7	0	Other	0	246	100	-9	-9	Other	100
80	-9	100	-8	No	100	164	-8	-7	0	Other	-8	247	0	-9	-9	Other	-9
81	100	0	-8	No	100	165	-9	-7	0	Other	-9	248	-7	-9	-9	Other	-9
82	0	0	-8	No	-7	166	100	-8	0	Other	100	249	-8	-9	-9	Other	-9
83	-7	0	-8	No	-7	167	0	-8	0	Other	-8	250	-9	-9	-9	Other	-9
84	-8	0	-8	No	-7												

Notes. Coding: positive value (represented by 100 in the table) stands for any strictly positive amount; -7 = did not have that occupation; -8 = no income in reference month; -9 = DK/NR; 0 = not applicable. The control variable takes value “No” (the household reported not relying on the income source) or “Other” (any other answer including “Yes,” DK, or NR). The column *E* reports the assignment to the reconstructed legacy aggregate under the rule of Section A.2: positive components dominate by sum; if no positive value is present and the control flag is “No,” $E = -7$; otherwise the negative-code hierarchy $-9 \succ -8 \succ -7$ propagates the most informative response code; the residual is 0.

OA-B Latent Factor Infrastructure

This online appendix provides the complete construction documentation for the latent-factor inputs summarized in print Section B.1. Section OA-B.1 documents the Housing Quality Index (ICH), which enters the seven-proxy measurement system as the source variable for the housing residual proxy. Section OA-B.2 documents the construction of each of the seven proxies, the identification battery, the correlation structure, and the missingness handling. Section OA-B.3 reports the full heterofactor estimation: technical settings, factor loadings, the external-consistency battery used for model selection, and the restricted-window and proxy-invariance robustness checks.

OA-B.1 The Housing Quality Index

The ICH is a Multiple Correspondence Analysis (MCA) summary of ten categorical housing indicators. It is constructed once over the full panel (2016Q4–2025Q3, $N = 1,840,332$) and used both as a regressor in the predictive stage and as the source variable for the housing-residual proxy of Section OA-B.2.

Indicators. Table OA-B.1 lists the ten housing indicators. Each is constructed from the raw EPH housing variables by grouping categories into substantively meaningful levels: structural quality (floor, roof), access to basic services (water, sanitation, cooking fuel), dwelling characteristics (type, number of rooms, garage, laundry room), and tenure security. Categories within each indicator are ordered from highest to lowest quality.

Table OA-B.1: Housing indicators in the ICH construction

Indicator	Categories	Construction rule
Housing type	house, apartment, other	Dwelling classification
Number of rooms	1, 2, 3, 4, 5+	Rooms ≥ 5 collapsed
Floor quality	high, medium, precarious	Material-based (tiles/wood vs. cement vs. dirt)
Roof quality	high solidity, medium, precarious	Two-factor: material $\times$ surface finish
Water access	indoor network, external	Network supply AND interior connection
Sanitation	optimal, deficit	Requires bathroom, inside, flush, sewer
Cooking fuel	network gas, bottled gas, precarious	Grid gas vs. bottled vs. other
Tenure	owner, tenant, occupant	Ownership vs. rental vs. informal
Garage	yes, no	Binary
Laundry room	yes, no	Binary

Estimation. Residual missing values in the ten indicators are imputed using a Random Forest algorithm (missRanger package, predictive mean matching with $k = 3$, global seed for reproducibility); missingness is negligible at the household level. The MCA is estimated with survey sampling weights (pondera) over the pooled panel, so the factor structure is fixed across time and changes in the ICH over time reflect genuine shifts in the distribution of housing indicators rather than re-estimation artefacts. The first-dimension MCA coordinates are rescaled to a 0–100 index by min-max normalization with sign inversion, so that higher values correspond to better housing quality:

$$\text{ICH}_i = 100 - \frac{c_i - \min(c)}{\max(c) - \min(c)} \times 100,$$

where c_i is the first-dimension MCA coordinate for household i .

OA-B.1.0.1 Distribution of the score. The score has mean 74.0 and standard deviation 16.8 over the working sample of 50,000 observations, with deciles ranging from 49.7 to 91.3. The mass is concentrated at relatively high values with a long lower tail, consistent with most Argentine urban households having at least basic housing infrastructure but a non-trivial minority displaying substantial structural deprivation.

MCA structure. Table OA-B.2 reports the explained variance by dimension. The first dimension explains 12.5% of the total inertia, well above the reference value of $1/(K - J) \approx 5.0\%$ that would obtain under independence between categories. The second dimension explains 9.7%; dimensions 3–10 form a flat plateau between 4.8% and 5.9% with no interpretable breaks, so the structure beyond the first two dimensions is diffuse. Only the first dimension is retained: it is the dominant axis by both the elbow criterion and the semantic ordering test described below, and additional dimensions would add complexity without a clear payoff in the downstream backcasting exercise.

Table OA-B.2: MCA eigenvalues and explained variance for the ICH

Dimension	Eigenvalue	% Variance	% Cumulative
Dim 1	0.250	12.507	12.507
Dim 2	0.194	9.712	22.219
Dim 3	0.117	5.857	28.076
Dim 4	0.106	5.305	33.381
Dim 5	0.103	5.154	38.535
Dim 6	0.101	5.074	43.609
Dim 7	0.100	5.011	48.619
Dim 8	0.100	4.981	53.600
Dim 9	0.098	4.898	58.498
Dim 10	0.096	4.790	63.288

Notes. In MCA with categorical variables, absolute explained variance per dimension is often modest. The relevant criterion is the relative dominance of the first dimension over subsequent ones, which supports using Dim 1 as the basis of the index.

Indicator contributions and semantic ordering. The contribution profile is dispersed across the ten indicators rather than concentrated in a single one. The five strongest contributors to Dim 1 — sanitation (18.0%), medium-quality flooring (15.4%), bottled-gas cooking (12.5%), precarious roofs (12.0%), and external water access (9.6%) — all exceed the reference contribution of $100/K \approx 4\%$ but none individually accounts for a dominant share. The indicators that vary most sharply across the quality spectrum are those that load most heavily on the first dimension, supporting the substantive interpretation. The category coordinates on Dim 1 sort in the substantively expected direction: better-quality categories (indoor water, optimal sanitation, network gas, high-solidity roof) lie at one extreme; precarious categories at the other. Sign indeterminacy of the MCA axis is resolved by the normalization above.

External validation. The ICH correlates positively with log real income at 0.280, a moderate value that confirms the index captures housing-quality variation related to but distinct from current income (a correlation close to unity would suggest the index is redundant; one near zero would question its economic content). Households without unmet basic needs (UBN) score 77.1 on the ICH on average, against 52.6 for households with UBN, showing that the index is closely related to broader material deprivation while being constructed from independent inputs. Geographic variation aligns with known regional patterns: Patagonia ranks highest on average (mean 81.1), followed by Cuyo (77.7), Pampeana (74.4), GBA (72.3), NOA (72.2), and NEA (66.6).

Temporal stability. The MCA is estimated on the pooled panel rather than quarter by quarter, so the factor structure is fixed across time. The quarterly mean of the ICH remains within a narrow band between 73 and 75 across the 36 quarters of the panel, with the standard deviation equally stable. Quarter-specific sample sizes vary (notably during the 2020 pandemic disruption), but these variations do not translate into shifts in the level or dispersion of the index, supporting the interpretation of the ICH as a structural household characteristic rather than a noisy cyclical indicator.

OA-B.2 The seven-proxy measurement system

The seven proxies are the empirical inputs to the heterofactor estimation. Their construction is designed so that each proxy draws its measurement error from an institutionally distinct mechanism within its factor block, satisfying the operational form of the conditional-independence requirement for factor identification (Kotlarski, 1967). Print Section B.1 provides a one-line summary of each proxy in Table B.1; this subsection reports the construction logic and design rationale in full.

B.2.1 Cognitive proxies. Four proxies load on θ_A .

School delay (cohort-standardized). Conditional on final educational attainment, completing each schooling level on time is interpreted as evidence of greater cognitive efficiency. Raw school delay — $d_i = \text{age}_i - \text{years educ.}_i - 6$, where 6 is the statutory entry age in Argentina — is contaminated by secular trends in educational access and by cohort-specific macroeconomic shocks (most notably the 2001 crisis). Standardization within birth cohort $c(i) = \text{survey year}_i - \text{age}_i$ removes these confounders:

$$\text{school_delay}_i = \frac{d_i - \bar{d}_{c(i)}}{s_{c(i)}}.$$

The remaining variation is interpreted as relative efficiency within cohort. The proxy loads negatively on θ_A . The measurement error is driven by idiosyncratic schooling shocks (illness, family disruption, grade repetition), institutionally distinct from the noise sources of the other three cognitive proxies.

Household educational climate. Defined as the mean years of completed education among adults aged 25 or older in the household, where 25 is chosen because formal education is typically completed by that point in Argentina. The data-generating mechanism differs from own schooling because it reflects the family learning environment rather than self-reported educational attainment — the measurement error comes from household composition

dynamics, not from individual credential reporting. Coverage is achieved through a three-level hierarchical fallback: if no eligible adult is in the household, the agglomerate-period mean is used; if that is also missing, the national-period mean substitutes. The final proxy has coverage above 99.9%.

Assortative matching. Under Becker (1973) assortative matching, individuals tend to form partnerships with others of similar latent ability. Spousal education is therefore a signal of latent skill that goes beyond own credentials. The measurement error arises from marriage-market frictions (timing, geography, preferences), institutionally orthogonal to the credentialing noise of school delay and the household-composition noise of educational climate. For individuals in couple households, the partner’s education is directly observed (approximately 67% of cases). For individuals without an observed partner, the proxy is imputed as the conditional mean by age group, sex, and region.

Normalized occupational qualification. Under a Roy-type assignment model with frictions, workers with higher θ_A have comparative advantage in more complex tasks. The CNO-based occupational qualification (a complexity score derived from the Argentine occupational classification) is used as a contemporaneous signal of productive capacity. The measurement error reflects job-search frictions and employer-side screening, not credential reporting. The high NA share (54.8%) is structural: only the economically active population with valid occupational history can be assigned a CNO qualification; non-EAP individuals receive NA. The proxy is z -standardized over the panel.

B.2.2 Socioemotional proxies. Three proxies load on θ_B .

Stability entropy. Traits associated with Conscientiousness and Emotional Stability are expected to manifest in more stable labour trajectories. The proxy is defined as inverted normalized Shannon entropy over three states (Employed, Unemployed, Inactive):

$$\text{stability}_i = 1 - \frac{H_i}{\log_2 3}, \quad H_i = - \sum_{k \in \{E, U, I\}} p_k \log_2 p_k,$$

where p_k is the fraction of observed waves in which individual i occupies state k . The proxy lies in $[0, 1]$: an individual always observed in the same state scores 1; one equally distributed across all three states scores 0. Three states (rather than four including formal/informal) are used because the formality classification is only available from 2024Q4 onward; including it would destroy coverage in the historical panel needed for backcasting. The measurement error arises from idiosyncratic labour-market shocks (layoffs, health events).

Housing residual. Individuals with greater patience or lower time discounting are expected to invest more in housing quality conditional on income. To purge the income channel, the proxy is the residual from

$$\text{ICH}_i = \beta_0 + \beta_1 \log(\text{income}_i + 1) + \beta_2 \text{age}_i + \beta_3 \text{age}_i^2 + \gamma' \text{region}_i + \delta' \text{educ}_i + \eta' \text{year}_i + u_i,$$

where the +1 shift on income retains zero-income households in estimation, the quadratic in age captures the life-cycle pattern of housing investment, region fixed effects absorb persistent geographic differences, education controls for the human-capital channel, and year fixed effects absorb secular trends. The R^2 of this regression is 0.230, indicating that the controls explain roughly a quarter of ICH variation — enough to purge the main confounders while preserving substantial residual variance attributable to unobserved preferences. Income is missing for 22% of the panel (a structural feature of the EPH); the residual is imputed at the median by agglomerate $\times$ year $\times$ quarter, with at least 266 donors per cell (median 1,123), and an imputation flag is carried forward.

Formal job-search intensity. Among the unemployed, the form of job search reveals Grit and internal locus of control. The proxy counts the number of active search methods used from a set of four: presenting oneself to firms, sending CVs or responding to advertisements, consulting employment agencies, and visiting job boards in person. Informal channels (personal contacts, neighbourhood notices) are excluded because they may reflect social capital and network density rather than personal agency. The proxy is coded NA for the non-unemployed, not zero: in an earlier specification with non-unemployed coded zero, approximately 95% of the sample shared the same value, causing θ_B to collapse onto a near-exact copy of the proxy ($R^2 = 99.2\%$). Setting non-unemployed to NA eliminates this dominance problem; the FIML estimation integrates over the missing data without imputation, while the other two socioemotional proxies anchor θ_B for the non-unemployed.

Descriptive statistics. Table OA-B.3 reports the moments and coverage of the seven proxies. Coverage differs sharply across variables because some proxies are structurally restricted to specific sub-populations.

Table OA-B.3: Descriptive statistics of the seven proxies

Proxy	Factor	Valid N	NA%	Mean	SD	Median	P25 / P75
School delay (cohort)	θ_A	1,448,945	21.3	−0.000	1.000	−0.061	−0.708 / 0.667
Educ. climate (hh)	θ_A	1,840,332	0.0	11.857	3.137	12.000	9.500 / 14.167
Assortative matching	θ_A	1,822,810	1.0	12.102	2.237	12.154	11.866 / 12.688
Occup. qualification	θ_A	831,189	54.8	0.000	1.000	0.046	0.046 / 0.046
Stability entropy	θ_B	1,610,403	12.5	0.883	0.245	1.000	1.000 / 1.000
Housing residual	θ_B	1,840,332	0.0	0.598	13.341	2.457	−3.983 / 8.362
Formal search	θ_B	57,921	96.9	1.145	0.902	1.000	1.000 / 2.000

Notes. Coverage differs across variables because some proxies are structurally restricted to specific sub-populations: occupational qualification is defined only for the economically active with occupational history; formal search is defined only for the unemployed. Educational climate, assortative matching, and the housing residual achieve full or near-full coverage through the hierarchical fallback or imputation procedures described in the construction details above.

Identification battery. Table OA-B.4 reports the five empirical pre-conditions checked for the seven-proxy system. Q1–Q2 verify that each proxy carries enough information and coverage to contribute to the likelihood. Q3–Q4 check that within each factor block no two proxies are nearly collinear (which would effectively reduce the system to a single measurement per factor). Q5 verifies that the cross-factor correlations remain close to zero, the empirical signature of factor separability. All five conditions pass.

Table OA-B.4: Identification checks for the seven-proxy system

Test	Observed value	Requirement	Result
Q1: Sufficient variance	min SD = 0.245	SD $\gg$ 0	PASS
Q2: Coverage	max NA = 54.8% (structural)	Adequate non-missing mass	PASS
Q3: No intra-cognitive dominance	max correlation = 0.541	Well below collinearity	PASS
Q4: No intra-socio dominance	max correlation = 0.016	Well below collinearity	PASS
Q5: Cross-factor orthogonality	max correlation = 0.166	Near-orthogonality	PASS

Notes. Q3–Q5 do not rely on fixed numerical thresholds; the requirement is that within-factor correlations remain far from unity (indicating distinct measurement mechanisms) and cross-factor correlations remain close to zero (indicating factor separability). The observed values satisfy both conditions by wide margins.

Cross-factor correlations. Table OA-B.5 reports the cross-block correlations underlying Q5. All values are below 0.17, confirming that the cognitive and socioemotional dimensions are empirically separable.

Table OA-B.5: Cross-factor correlations

θ_A proxy	Stability entropy	Housing residual	Formal search
School delay (cohort)	−0.045	−0.046	−0.097
Educ. climate (hh)	0.073	0.128	0.122
Assortative matching	0.030	0.048	0.129
Qualification (norm.)	0.166	0.068	0.043

Temporal stability. Quarterly means of the seven proxies evolve smoothly over 2016Q4–2025Q3 without breaks. The cognitive proxies display gradual movements consistent with secular trends in educational attainment and labour-market composition. The socioemotional proxies are also broadly stable, although formal search is naturally more volatile because it is defined only for the unemployed.

OA-B.3 Heterofactor estimation and robustness

The heterofactor model recovers two latent factors from the seven-proxy system through a measurement equation with restricted cross-loadings. Print Section B.1 states the equation and the model-selection rationale; this subsection reports the full estimation results, the external-consistency battery, and the robustness analysis.

Technical settings. The model is estimated by Full-Information Maximum Likelihood with the integral over (θ_A, θ_B) evaluated by Gauss–Hermite quadrature. The configuration is fixed across both candidate models. Table OA-B.6 reports the parameters.

Table OA-B.6: Heterofactor estimation: technical settings

Parameter	Value
N_MAX_MLE (stratified subsample for parameter estimation)	20,000
GH quadrature grid	9×9 (81 nodes)
Convergence tolerance	10^{-6}
N threads	16
GPU device	cuda
FIML over missing proxies	TRUE
Random seed	123

Notes. N_MAX_MLE is the stratified subsample used for parameter estimation; the full scoring universe of 1,840,332 observations is used for posterior factor scores via empirical Bayes posterior means. FIML handles proxy-level missingness natively: when proxy j is missing for individual i , the corresponding term is omitted from the individual likelihood contribution, avoiding the severe coverage loss that complete-case estimation would impose.

Model A vs Model B. Two specifications are estimated on the same proxy system, differing only in the observable controls X_i in the measurement equation. Model A is parsimonious: sex, a quadratic in age, and region fixed effects, totalling 77 free parameters. Model B adds log-ICH and agglomerate fixed effects, increasing the parameter count to 301. Table OA-B.7 reports the comparison.

Table OA-B.7: Heterofactor: technical comparison between Model A and Model B

Metric	Model A	Model B
Free parameters	77	301
Log-likelihood	-195485.9	-199632.7
Convergence	Yes	Yes
Maximum iterations	150	1,200
Estimation time (minutes)	13.4	115.5

The likelihood-ratio statistic is -8293.48 on 224 degrees of freedom with $p < 2.2 \times 10^{-16}$, which would mechanically select Model B. Two reasons argue against this: with a sample of 1.4 million observations, even trivially small improvements in fit generate overwhelmingly significant LR statistics — the test has power to detect differences that are statistically real but economically negligible; and the latent factors are not endpoints but inputs to the downstream LASSO and backcasting stages, where their economic gradients matter more than the within-sample likelihood of the measurement model. The relevant criterion is the external consistency of the recovered factors against variables not used in estimation.

Factor loadings. Table OA-B.8 reports the factor loadings under both specifications. The proxy partition across factors is preserved in both, but the cognitive-factor magnitudes shift markedly under Model B, with the socioemotional-side loadings becoming numerically extreme (notably housing residual at 44.1 and formal search at 6.5 under Model B), a sign of the absorption of variance by the additional control set rather than of substantively different measurement.

Table OA-B.8: Factor loadings and idiosyncratic standard deviations

Proxy	Factor	Model A			Model B		
		α_A	α_B	σ_ε	α_A	α_B	σ_ε
School delay (cohort)	θ_A	-0.518	0.000	0.945	-0.449	0.000	0.942
Educ. climate (hh)	θ_A	1.029	0.000	0.829	0.925	0.000	0.801
Assortative matching	θ_A	1.915	0.000	0.317	1.897	0.000	0.315
Occup. qualification	θ_A	0.441	0.000	0.946	0.337	0.000	0.929
Stability entropy	θ_B	0.000	0.055	1.031	0.000	-9.024	1.022
Housing residual	θ_B	0.000	-0.038	0.958	0.000	44.106	0.577
Formal search	θ_B	0.000	-1.067	0.018	0.000	6.480	0.905

Notes. Cross-loadings are restricted to zero in line with the proxy-design assignment. The partition is preserved in both models; the magnitude differences in Model B are driven by the absorption of variance into the additional control set rather than by substantively different measurement.

External-consistency battery. Table OA-B.9 reports the eight external-consistency checks used for model selection. Each check evaluates a directional relationship between the recovered factors and a variable not used in estimation: Cohen's d between formal and informal workers on each factor (the factor should be higher for formals); the LPM coefficient on each factor when predicting formality (should be positive); the incremental R^2 from adding the two factors to the demographic regression (should be positive); the correlation of each factor with the income decile (cognitive should be positive but not unity); and the relationship of θ_A with the ICH score and household educational climate (should be positive).

Table OA-B.9: External consistency checks for Model A and Model B

Metric	Model A	Model B
d Cohen θ_A (formal – informal)	+0.313	−0.313
d Cohen θ_B (formal – informal)	+0.485	+0.485
LPM coefficient on θ_A (sign)	+2.249	+2.249
ΔR^2 from adding (θ_A, θ_B) to demographic LPM	+0.060	+0.060
Correlation: income decile vs. θ_A	+0.795	+0.795
Correlation: income decile vs. θ_B	+0.912	+0.912
ICH score $\sim \theta_A$ (sign)	−0.091	−0.091
Educ. climate $\sim \theta_A$ (sign)	−0.014	−0.014
Criteria satisfied	5/8	4/8

Notes. Model A is retained because it yields a cognitive factor with the correct directional interpretation in the formality and income comparisons. Model B produces a structurally inverted θ_A : informal workers rank above formal ones on the cognitive factor, contradicting the theoretical assignment. The richer control set in Model B absorbs variation that should flow through the latent factor. Because the factors are inputs to the predictive stage, directional coherence is the binding criterion for selection.

The decisive contrast is the sign of Cohen’s d on θ_A between formal and informal workers: +0.313 under Model A but −0.313 under Model B. In Model B, informal workers rank *higher* than formal workers on the cognitive factor, contradicting the theoretical assignment. This is not a sign-indeterminacy issue (factor signs are consistently fixed by the loading-direction convention); it is a substantive inversion. The richer control set of Model B absorbs precisely the variation that should flow through θ_A , leaving the factor to capture residual variation that is anti-correlated with formality. This inversion would propagate into the predictive stage: a model trained with inverted θ_A would assign higher formality probabilities to individuals with lower cognitive-skill scores, generating a historical series whose interpretation contradicts the theoretical framework.

Restricted-window robustness. A natural concern is whether estimating the heterofactor on the full panel 2016Q4–2025Q3 introduces look-ahead information from the four post-redesign quarters in which formality is directly observed. To address this, we re-estimate Model A on the restricted panel 2016Q4–2024Q3, excluding the post-redesign quarters. Table OA-B.10 compares the two sets of factor scores on the matched sample.

Table OA-B.10: Robustness: full-sample versus restricted factor scores (Model A)

Factor	Pearson r	Spearman ρ	RMSE	N
θ_A (Cognitive)	0.86	0.537	0.116	1,338,874
θ_B (Socioemotional)	0.9998	0.9981	0.012	1,338,874

Notes. Comparison of posterior factor scores between the full-sample Model A (2016Q4–2025Q3) and the restricted Model A (2016Q4–2024Q3). Both runs use the same proxy system, loading structure, and BFGS optimiser. The matched sample contains all individuals with factor scores in both runs.

The socioemotional factor is essentially invariant to the estimation window. The cognitive factor exhibits a moderate Pearson correlation but a negative Spearman — a divergence diagnostic of a Gauss–Hermite quadrature artefact: in the full-sample model, the empirical Bayes posterior mean concentrates approximately 2% of individuals at the extreme quadrature nodes ($\pm 1.02\sigma$ and $\pm 2.08\sigma$), producing a bimodal score distribution. In the restricted model the distribution is unimodal and narrower. A cross-decile analysis of the matched sample confirms that the apparent rank reversal is driven entirely by these node-boundary cases; for the central 90% of the distribution the ordinal structure is preserved. The negative Spearman is therefore a quantization effect of the 9×9 grid at the extreme tails, not substantive instability of the latent ordering.

Downstream impact on the bridged series. A separate question is whether the moderate rescaling of θ_A propagates to the final backcasted series. Two counterfactual scenarios are evaluated. Scenario A replaces the full-sample θ_A and θ_B with the restricted-window scores and re-predicts using the same post-LASSO GLM and calibrated threshold. Scenario B sets θ_A to its training-sample mean (effectively removing its contribution) and re-predicts. Table OA-B.11 reports the comparison against the headline series.

Table OA-B.11: Downstream impact on the hybrid formality series: restricted factors and removal of θ_A

Scenario	Pearson r	RMSE (pp)	Max $ \Delta $ (pp)
A: Restricted factors	0.526	10.12	12.37
B: Without θ_A	0.907	1.27	2.32

Notes. Baseline is the hybrid formality series using the full-sample factor scores. Correlations and RMSE computed over 32 quarterly rates. Both scenarios use the same post-LASSO GLM coefficients and calibrated threshold. In the hybrid series, directly observed formality classifications (2024Q4–2025Q3) are preserved; only the predicted block is affected by the factor substitution.

The two scenarios produce nearly identical deviations from the baseline. This is informative: the difference between the restricted-factor and baseline series is driven primarily by the cardinal level shift in θ_A (mean shift translating into a roughly +4 pp upward push on predicted probabilities via the GLM coefficient $\hat{\beta}_{\theta_A} = -0.371$),

rather than by any change in the temporal pattern. The shape of the predicted block — trends, turning points, cyclical dynamics — is highly stable. The qualitative conclusions of the paper (the downward shift relative to the legacy proxy, the pandemic disruption, the post-2022 stabilization) are entirely robust to the choice of estimation window for the heterofactor.

Proxy invariance across the redesign boundary. A final check verifies that the seven proxy variables are themselves measured consistently across the 2025 redesign — a prerequisite for full-panel factor scoring. The redesign modified the labour-market module but did not alter the demographic, educational, or housing blocks from which the proxies are constructed. Table OA-B.12 reports formal distributional comparisons (Cohen’s d and Kolmogorov–Smirnov statistics) between the pre-redesign (2016Q4–2024Q3) and post-redesign (2024Q4–2025Q3) periods.

Table OA-B.12: Measurement invariance of the seven heterofactor proxies

Proxy	$\bar{x}_{\text{pre}}$	$\bar{x}_{\text{post}}$	Cohen’s d	KS D (full)	KS D (boundary)
ICH score	75.075	75.283	0.013	0.012	0.009
Housing residual	0.120	−0.267	−0.032	0.077	0.072
School delay (cohort)	−0.126	0.966	1.251	0.452	0.013
Educ. climate (hh)	12.330	12.592	0.087	0.050	0.014
Assortative matching	12.525	12.642	0.048	0.023	0.006
Occup. qualification	0.026	0.075	0.050	0.033	0.010
Stability entropy	0.890	0.919	0.132	0.051	0.021

Notes. Cohen’s d is the standardized mean difference (pooled SD). “Full” compares the entire pre-redesign occupied sample ($\sim 700\text{k}$) against the post-redesign sample ($\sim 62\text{k}$); the large- N KS test rejects for any non-trivial shift. “Boundary” compares only 2024Q3 vs. 2024Q4 (the quarters immediately flanking the redesign), isolating the redesign transition from secular trends.

6 of 7 proxies show negligible effect sizes ($|d| < 0.20$) and boundary KS statistics below 0.08, confirming measurement invariance across the redesign. The exception is school delay, which displays a large $d = 1.25$ in the full comparison but a boundary KS of only 0.013 — indicating that the shift reflects a secular trend in educational attainment over 2016–2025 rather than a redesign-induced discontinuity. None of the seven proxies relies on the labour-market variables modified by the 2025 redesign; all are constructed from the demographic, educational, and housing modules, which were unchanged.

OA-C Predictive Model Diagnostics

This online appendix expands the predictive-stage results summarized in print Section B.4. Section OA-C.1 reports LASSO selection patterns by predictor block. Section OA-C.2 reports the headline GLM average marginal effects with discussion of the qualitative similarity across LPM, GLM, and SLS specifications. Section OA-C.3 reports the Random Forest benchmark referenced in print Section B.5. Section OA-C.4 documents the agreement between the dual-path reconstruction and the official INDEC derived field. Per-model coefficient tables (96–97 selected variables each, with cluster-robust standard errors and p -values) and the complete table of GLM AMEs across all selected predictors are generated by the replication package and follow the structure described below.

OA-C.1 LASSO selection patterns

Of the 112 candidate columns after dummy expansion, the LPM retains 97, the GLM retains 96, and the SLS retains 97 variables at $\lambda_{1\text{se}}$, of which 95 survive the post-trimming OLS. The three models select essentially overlapping predictor sets: pairwise overlap of selected variables exceeds 95% in every comparison. The latent factors θ_A and θ_B enter every specification by construction (zero LASSO penalty). Table OA-C.1 reports approximate selection counts by predictor block.

Table OA-C.1: Approximate LASSO selection counts by predictor block

Predictor block	Candidates	LPM	GLM	SLS
Demographics (sex, age, region, agglomerate)	~ 40	~ 38	~ 36	~ 38
Education and human capital	~ 12	~ 10	~ 10	~ 10
Labour status (occupational category, sector)	~ 24	~ 20	~ 22	~ 21
Job characteristics (hours, tenure, firm size)	~ 12	~ 11	~ 11	~ 11
Household composition	~ 10	~ 8	~ 8	~ 8
Longitudinal proxies and ICH	7	7	7	7
Latent factors θ_A, θ_B (forced)	2	2	2	2
Total	112	97	96	97

Notes. Counts are approximate by predictor block and reflect the typical pattern at $\lambda_{1\text{se}}$. Variation across models in any single block does not exceed two variables. The seven longitudinal proxies plus the ICH score are retained by all three models, indicating high cross-validated informativeness.

Bootstrap stability. Variables retained in 80% or more of 200 household-clustered bootstrap resamples form the core stable predictor set: the latent factors θ_A and θ_B , log hours worked, log job tenure, normalized

occupational qualification, household educational climate, the housing residual, the ICH score, indicators for tertiary or higher education and for public-sector employment, and broad sectoral dummies (industry, commerce-services, public administration). The remaining selected variables exhibit bootstrap selection rates between 30% and 80% and represent secondary refinements; their inclusion improves cross-validated fit marginally but is sensitive to the resample. The pruned specification of print Section B.5 retains only the stable set and produces a hybrid series correlated at $r = 0.974$ with the headline.

OA-C.2 GLM average marginal effects

Table OA-C.2 reports headline GLM average marginal effects (in percentage points), computed as the population-weighted average of partial derivatives evaluated at observed covariate values. The reference categories are reported in the table notes. The complete table including all 96 selected predictors, cluster-robust standard errors, confidence intervals, and p -values is generated by the R replication pipeline and follows the same block structure.

Table OA-C.2: Headline GLM average marginal effects on the probability of formality

Predictor	AME (pp)	p -value
<i>Latent factors (per standard deviation)</i>		
θ_A (cognitive)	+7.07	< 0.001
θ_B (socioemotional)	+0.83	0.143
<i>Occupational category (ref: wage employee)</i>		
Employer	+2.82	< 0.001
Self-employed	−18.75	< 0.001
Family worker	−56.76	< 0.001
<i>Sector of activity (ref: public administration)</i>		
Teaching	−6.96	< 0.001
Health	−13.27	< 0.001
Industry	−18.47	< 0.001
Commerce	−22.20	< 0.001
Personal and community services	−29.78	< 0.001
Construction	−37.66	< 0.001
Domestic service	−42.65	< 0.001
<i>Education (ref: primary complete)</i>		
University complete	+12.12	< 0.001
Tertiary complete	+10.79	< 0.001
Secondary complete	+5.94	< 0.001
Primary incomplete	−3.94	< 0.001
<i>Household and longitudinal proxies</i>		
ICH score	+9.14	< 0.001
Tenure (per year)	+2.71	< 0.001
Educational climate of household	+1.87	< 0.001
<i>Demography</i>		
Age (centred)	+25.91	< 0.001
Age squared	−22.22	< 0.001
Born abroad (other country)	−9.63	< 0.001
Women	−2.14	< 0.001

Notes. Average marginal effects from the preferred GLM (logit), expressed in percentage points and computed as population-weighted means of partial derivatives at observed covariate values. Standard errors and p -values are computed by the delta method. The full table covering all selected predictors and the residual urban-agglomerate effects is reported in print Table ??; the comprehensive ranked list including household-income-source dummies, kinship indicators, and small-cell categories is in print Appendix F. Rows shown here are a curated subset chosen for substantive readability rather than the largest absolute effects: the largest individual AMEs are the family-worker, self-employed, and sectoral coefficients; the smallest are the latent-factor and longitudinal-proxy coefficients, which capture residual variation conditional on observables. Reference categories are made explicit in each block; firm-size and public-vs-private dummies do not appear because firm size (`tamano_empresa`) was not retained at λ_{1se} and the public-sector indicator is implicit in the omitted sector category (public administration).

Substantive interpretation. The AME profile is consistent with the established formality literature for Argentina. The strongest individual contributors are occupational and sectoral: family workers face a −56.76 pp gap relative to wage employees, the self-employed face a −18.75 pp gap, while employers carry a small positive effect of +2.82 pp; sector coefficients range from −6.96 pp for teaching to −42.65 pp for domestic service relative to public administration. Educational attainment shows a strong monotone gradient: university complete contributes +12.12 pp and tertiary complete +10.79 pp relative to primary complete, while primary incomplete is −3.94 pp. Household and longitudinal indicators contribute as expected: a one-point increase in ICH adds +9.14 pp, an additional year of job tenure +2.71 pp, and household educational climate +1.87 pp per unit. The latent factors enter with the expected signs but with markedly different magnitudes. The cognitive factor θ_A contributes +7.07 pp per standard deviation ($p < 0.001$), capturing a residual cognitive-skill premium net of explicit education credentials and observable proxies. The socioemotional factor θ_B enters at +0.83 pp per standard deviation but is not statistically distinguishable from zero ($p = 0.143$). This pattern is consistent with the heterofactor robustness analysis (online Section OA-B.3): θ_B supports the historical bridge primarily by transmitting persistence through the LASSO-selected longitudinal proxies (school delay, educational climate, tenure, housing residual) that load on it, rather than by exerting a direct marginal effect at the GLM stage. The qualitative ordering of the AME profile—occupational and sectoral coefficients dominating, education and longitudinal proxies as second-order, latent factors third-order—is consistent with the linear benchmark reported in print Figure ?? and is therefore not an artefact of the nonlinear specification.

Qualitative comparison across LPM, GLM, and SLS. The signs and relative magnitudes of the headline coefficients are preserved across the three model specifications, which is expected given near-identical predictor selection (Section OA-C.1). LPM and SLS coefficients are linear and read directly as percentage-point changes in the formality probability; comparison with the GLM AMEs of the table above is therefore direct, with the LPM coefficients typically falling within ± 0.3 pp of the corresponding GLM AME and the SLS coefficients showing slight compression toward zero (a consequence of the iterative trimming of extreme cases). The qualitative pattern — positive contributions from the latent factors, education, firm size, and public-sector employment; negative from self-employment, domestic work, and the agrarian sector — is identical across the three specifications. The decisive differences across models lie not in coefficient pattern but in calibration and admissibility (print Section B.4).

OA-C.3 Random Forest benchmark

A 500-tree Random Forest is fitted on the same predictor set as the GLM, with the same training-test split, sampling weights, and household-clustered cross-validation folds. Hyperparameters are tuned over the grid $mtry \in \{8, 12, 16\}$, minimum node size $\in \{5, 10, 20\}$ on the training set; the selected configuration is $mtry = 12$, minimum node size = 10. Table OA-C.3 reports the comparison.

Table OA-C.3: Random Forest versus GLM: predictive performance on the test fold

Metric	GLM (logit)	Random Forest
AUC-ROC	0.990	0.974
Brier score	0.039	0.067
Expected calibration error (%)	0.4	4.9
Calibration slope	1.031	1.665
Predictions outside $[0, 1]$	0	0

Notes. All metrics are computed on the held-out test fold of the four-quarter overlap window. Both specifications produce predictions bounded in $[0, 1]$ by construction (the Random Forest by majority voting, the GLM by the inverse logit). Calibration slope below 1 indicates systematic compression of predicted probabilities relative to observed rates.

The Random Forest matches the GLM on discrimination but is materially worse on calibration. Two consequences follow. First, the GLM is not dominated by a more flexible specification on this prediction problem: the linear log-odds structure with the latent-factor inputs is competitive with a non-parametric ensemble on AUC-ROC while delivering substantially better-calibrated probabilities. Second, deploying the RF for backcasting would require an additional probability-calibration step (Platt scaling or isotonic regression) before threshold application, adding methodological steps without a discrimination payoff. The GLM’s combination of competitive ranking, proper calibration, bounded predictions, and parametric transparency makes it the operationally preferred specification.

OA-C.4 Agreement with the INDEC official derived field

The 2024Q4 EPH microdata include a derived variable, `formalidad_empleo`, constructed by INDEC from the underlying primitives that classifies each occupied worker as formal or informal under the dual-path concept (INDEC, 2025). The dual-path reconstruction documented in print Section A.3 produces an agreement of 100% with this field on the employee subuniverse, where both rules apply `desc_jubilatorio_asalariado` (constructed from PP07H) as the determining variable. There is no scope for divergence: the rules are identical on this subuniverse.

For non-wage workers, exact agreement with the official derived field is not the relevant comparison. The INDEC convenience field carries a residual “DK/NR” category that the bridge resolves through the underlying primitives, on the principle that DK/NR is not informationally equivalent to “No” but reflects a distinct response state about a question that may itself have skip-routing complications. The bridge therefore produces a binary classification (formal or informal) where the convenience field has three states (formal, informal, DK), and the comparison reduces to whether the bridge’s resolution of the DK cases is empirically defensible. In the four overlap quarters, DK/NR represents below 4% of non-wage workers; the bridge classifies the majority of these as informal (consistent with the prior that affirmative formalization signals are observed when present), and the resulting non-wage formality rates align with administrative *monotributo* enrolment counts (Appendix D).

Pre-2024 historical period. For the historical (pre-2024Q4) period there is no INDEC official non-wage formality derived field with which to compare, because the underlying primitives required for the dual-path classification (PP05I, PP05B3/PP05K, PP06E1) were not collected. The bridge fills exactly this gap. Validation in the historical universe therefore relies on the indirect coherence checks reported in the validation appendices: 100% agreement on employees with the legacy proxy, aggregate co-movement with SIPA in changes (Appendix D), and subgroup stability of the legacy-bridged gap (Section C.4). No alternative validation path is available, and this constraint should be acknowledged when interpreting the historical bridged series.

OA-D Backcasting and Hybrid Diagnostics

This online appendix expands the backcasting and hybrid-construction diagnostics summarized in print Section C. Section OA-D.1 reports admissibility patterns and qualitative behaviour of the three predictive specifications across the three nested universes. Section OA-D.2 compares Youden and calibration thresholds across models and reports the threshold-sensitivity diagnostics. Section OA-D.3 reports cross-model agreement and the GLM confusion matrix decomposed by occupational segment. Section OA-D.4 documents the stability of the predicted block's occupational composition. Section OA-D.5 reports potential-formality margins for the unemployed and inactive sub-universes.

OA-D.1 Per-model behaviour across the three universes

The three predictive specifications — LPM, GLM, SLS — are applied to three nested universes ordered by decreasing strength of identification: the occupied population (ESTADO==1), the economically active population (occupied and unemployed), and the broader 18–60 universe (occupied, unemployed, and inactive). Print Section C.1 discusses the qualitative pattern; this subsection reports the admissibility profile that drives the GLM-vs-LPM-vs-SLS comparison, together with figure references to the per-model quarterly series.

Admissibility profile. Table OA-D.1 reports the share of raw predictions falling outside the unit interval before clipping, by model and universe. The GLM is bounded in $[0, 1]$ by construction; the LPM and SLS produce inadmissible predictions whose share grows monotonically as the universe expands beyond the redesign-era occupied core on which the models were estimated.

Table OA-D.1: Predictive-rule admissibility across the three universes

Universe	LPM (% outside $[0, 1]$)	GLM (% outside $[0, 1]$)	SLS (% outside $[0, 1]$)
Test sample (overlap, occupied)	10.50	0.00	0.00
Historical occupied universe (32 pre-redesign Q)	9.1	0.00	21.9
Unemployed extension	~81	0.00	41.1
Inactive 18–60 extension	≥ 90	0.00	≥ 55

Notes. The share of raw model predictions falling outside the unit interval before clipping. The GLM produces zero inadmissible predictions in every domain. The LPM and SLS are estimated on the redesign-era occupied core (the only universe where formality is directly observed); when projected onto the unemployed and inactive 18–60 sub-populations, both produce a rapidly increasing share of inadmissible predictions because these populations lie partly outside the support of the training sample. SLS in-sample trimming improves admissibility inside the converged $\hat{\kappa}_\gamma$ region but does not transfer.

Quarterly series, occupied universe. In the four overlap quarters where formality is directly observed for the entire occupied population, all three models reproduce the directly observed quarterly rates within the calibration deltas reported in Table OA-D.2 below. In the historical occupied universe (the 32 pre-redesign quarters of the panel), the three hybrid series differ by less than 1.0 pp in any quarter; pairwise correlations exceed 0.99 across the panel (Table OA-D.3). The three series are visually indistinguishable at the quarterly resolution of the bridged index. Per-model series are in figures 08c_reporte_hibrido_LPM_s1a_01.pdf, 08c_reporte_hibrido_GLM_s1a_01.pdf, and 08c_reporte_hibrido_SLS_s1a_01.pdf.

Quarterly series, EAP extension. Extending to the unemployed introduces meaningful quantitative differences. The unemployed retain labour-market histories (job tenure prior to unemployment, sector of last job) and latent-factor scores, but their formality status is counterfactual: the model predicts the probability that the individual would be classified as formal if observed in employment. Approximately 81% of LPM raw predictions in this extension fall outside $[0, 1]$ before clipping, and 41.1% of SLS predictions remain inadmissible because the unemployed lie partly outside the support of the converged $\hat{\kappa}_\gamma$ trimming sample. Per-model EAP-level series are in figure 09b_comp_retropredictivo_series_pea_04.pdf; the GLM produces a population-weighted formality probability around 0.41–0.43 across the panel, with quarter-to-quarter variation of ± 0.04 , reflecting the cyclical sensitivity of the unemployed sub-population.

Quarterly series, broader 18–60 universe. The third extension adds the inactive 18–60 population. Labour-market variables (sector, hours, tenure, firm size) are imputed at the training-set mode, so the prediction relies primarily on demographics, household composition, and the latent factors. The inactive-potential-formal block contains roughly one quarter of the panel and receives a predicted formal rate around 32.3% under the calibration threshold. Per-model series for this universe are in figure 09b_comp_retropredictivo_series_edad_05.pdf. Rate dispersion across LPM, GLM, and SLS in this extension reaches 4–6 pp by 2024, the highest of the three universes, reflecting the dominance of model assumptions where covariate information is sparsest. Users interested in the inactive 18–60 group should treat the predicted rates as model-implied counterfactuals rather than as population aggregates of directly observed phenomena.

OA-D.2 Threshold sensitivity

Predicted probabilities are converted to binary classifications using a calibration threshold rather than the Youden threshold. Print Section B.4 discusses the rationale: the calibration threshold is selected to minimize quarterly aggregate-rate discrepancies, the binding criterion for backcasting, while the Youden threshold maximizes individual-level classification accuracy on the test fold. Table OA-D.2 reports both thresholds for each model and the maximum quarter-level aggregate-rate discrepancy under each.

Table OA-D.2: Youden vs calibration thresholds and quarter-level discrepancies

Model	Youden threshold	Calibration threshold	Max delta, Youden (pp)	Max delta, calibration (pp)
LPM	0.5386	0.540	2.4	1.06
GLM	0.5381	0.545	1.8	1.39
SLS	0.5720	0.530	3.1	3.91

Notes. The Youden threshold maximizes sensitivity + specificity – 1 on the test fold and is designed for individual-level classification. The calibration threshold is selected to minimize the maximum absolute discrepancy between predicted and observed quarterly aggregate rates across the four overlap quarters. “Max delta” columns report the maximum quarter-level absolute discrepancy in the overlap window under each threshold. Threshold-sensitivity figures by model are in 09c_comp_hibrido_GLM_s2a_comp_03.pdf and the analogous LPM and SLS files.

The Youden threshold introduces a systematic level shift in aggregate rates of approximately 0.6–1.3 pp across the three models, and is therefore inappropriate for the bridge despite its individual-level optimality. Threshold sensitivity is largest for the SLS, where iterative trimming compresses predicted probabilities into a narrow range, making aggregate rates highly sensitive to the threshold cutoff.

Probability aggregation as alternative. A separate sensitivity exercise compares the GLM hybrid series under the calibration threshold against direct probability aggregation, in which the quarterly rate is computed as the mean predicted probability rather than the share of binary classifications above the threshold. The maximum quarter-level discrepancy between the two aggregation schemes is below 0.1 pp throughout the panel under the GLM. This confirms that the threshold choice is empirically secondary to the calibration of the model itself once the calibration threshold is adopted; the GLM’s calibration is sufficiently good that thresholding does not introduce material aggregate-rate distortion. The same comparison under the LPM and SLS shows discrepancies of approximately 0.4 pp and 0.7 pp respectively, reflecting their poorer calibration.

OA-D.3 Cross-model agreement and confusion matrices

Cross-model agreement on the hybrid series. Table OA-D.3 reports pairwise agreement metrics for the three hybrid series across the 36-quarter panel. The hybrid combining rule of print Section C.2 preserves observed classifications wherever the survey provides them; differences across hybrid series therefore reflect only differences in the predicted block (31.4%% of the panel).

Table OA-D.3: Cross-model agreement on the hybrid formality series

Pair	Pearson r (quarterly rates)	RMSE (pp)
LPM hybrid vs GLM hybrid	0.9988	0.41
LPM hybrid vs SLS hybrid	0.9938	0.39
GLM hybrid vs SLS hybrid	0.9951	0.27

Notes. Pairwise comparisons of quarterly hybrid formality rates across the 36 quarters of the panel (2016Q4–2025Q3). Pearson correlations exceed 0.99 in all pairs; RMSEs are below 0.5 pp. The three hybrid series are essentially indistinguishable at the level of quarterly aggregates. This is a stronger result than the pure-backcasting comparison: when observed information is reintroduced through the hybrid combining rule, the residual differences across predictive families collapse, confirming that the choice of model specification matters for individual-level prediction (where calibration and admissibility differ) far more than for aggregate-rate construction.

Internal agreement on the predicted block. Within the four overlap quarters, ground truth is available for the predicted block via the dual-path classification. The GLM produces global agreement of 79.42% with the directly observed classification, with 84.47% on the formal class and 70.01% on the informal class. The pattern (higher agreement for the formal class than the informal one) holds in the LPM and SLS as well, reflecting the more diffuse distribution of unit-level signals across the three formality indicators in the informal sub-population.

GLM confusion matrices in the overlap. Table OA-D.4 decomposes the GLM confusion matrix by occupational segment.

Table OA-D.4: GLM confusion matrices in the overlap window, by occupational segment

Segment	N	Accuracy	Precision	Recall	F1
Employees (CAT_OCUP = 3)	59,100	0.791	0.823	0.853	0.838
Independent (CAT_OCUP ∈ {1, 2})	22,128	0.784	0.777	0.620	0.689
Overall (occupied)	81,228	0.789	—	—	—

Notes. Confusion matrices for the GLM hybrid in the four overlap quarters (2024Q4–2025Q3, pooled). Metrics are unweighted within each segment to preserve cell-level interpretability. Overall expected calibration error is 2.64 pp. Lower recall on independent workers reflects the more diffuse distribution of unit-level signals across the three formality indicators (Section A.3); the reconstruction is conservative on the formal margin for non-wage workers, the safer direction for backcasting since false positives on the formal side would inflate the historical series.

OA-D.4 Occupational composition of the predicted block

A natural concern with predictive backcasting is whether the predicted block’s composition shifts over time, in which case quarter-to-quarter changes in the bridged series might reflect compositional change rather than substantive variation in formalization. Print Figure 2 shows that the predicted share remains close to 31% across the 36 quarters of the panel. Table OA-D.5 reports the breakdown of the predicted block by occupational category.

Table OA-D.5: Composition of the predicted block by occupational category

Occupational category	Predicted-block share (%)	Within-category share (%)
Employer (CAT_OCUP = 1)	11.4	100.0 (predicted)
Self-employed (CAT_OCUP = 2)	75.6	100.0 (predicted)
Family worker (CAT_OCUP = 4)	13.0	100.0 (predicted)
Total predicted block	100.0	31.4 of total panel

Notes. Composition of the predicted block in the GLM hybrid, averaged over the 36 quarters of the panel. The within-category share is 100% for each non-wage category by construction, since employees in pre-redesign quarters retain their legacy classification and enter the observed block. The predicted-block share is the proportion of predicted observations falling in each occupational category. The dominance of self-employment (75.6%) reflects the structure of non-wage employment in Argentine urban labour markets.

The composition of the predicted block is stable over time within ± 1.5 pp for each category across the panel. This stability is structural: it reflects the slowly evolving share of self-employment in the Argentine labour market over 2016–2025, rather than a feature of the prediction algorithm. The implication is that quarter-to-quarter variation in the bridged formality series can be interpreted as variation in formalization within a stable composition, rather than as the joint product of compositional change and rate change. Per-model composition diagnostics are in 09c_comp_hibrido_tasa_pred_bloque_07.pdf.

OA-D.5 Potential formality among the unemployed and inactive

The bridge produces predicted formality probabilities for the unemployed and the inactive 18–60 sub-populations as a by-product of the three-step funnel (print Section C.1). These predictions are not used in the headline bridged series, which is anchored on the occupied universe, but they support a separate analysis of latent formal-employment potential.

Unemployed. Among the unemployed, the GLM predicts a population-weighted formality probability of approximately 0.41 under the calibration threshold, with quarter-to-quarter variation of ± 0.04 . This counterfactual reflects what the unemployed would be classified as if observed in employment, given their labour-market histories, latent-factor scores, and demographic characteristics. The interpretation is bounded: predictions for unemployed individuals lying far from the support of the overlap-window training sample (very long unemployment spells, atypical labour-market trajectories) carry larger model-implied uncertainty than the headline bootstrap bands of print Figure 4 convey, since the bands propagate LASSO sampling uncertainty within the overlap but not transportability uncertainty to atypical sub-populations.

Inactive 18–60. The inactive-potential-formal block produces a predicted rate of 32.3% under the calibration threshold. Labour-market variables are imputed at the training-set mode (sector = commerce-services, hours = median, tenure = median), so the prediction relies primarily on demographics, household composition, and the latent factors. The rate should be interpreted as a counterfactual scenario: the formality probability that the average inactive 18–60 individual would receive if entering employment with mode-imputed job characteristics. It is not a population aggregate of any directly observed phenomenon.

Implications for users. Researchers using the bridged formality series should restrict aggregations to the occupied universe except when the counterfactual interpretation of the predicted-formality margin among the unemployed and inactive is the explicit object of analysis. The headline series of print Table 5 and Figure 3 are constructed from the occupied universe, where the bridge is anchored on observed and reconstructed classifications and where the prediction component is structurally bounded by the share of non-wage workers (Table 3).

OA-E SIPA Validation Extras

This online appendix expands the external-validation analysis summarized in print Section D. Section OA-E.1 reports per-segment series for total, employee, and independent registered employment. Section OA-E.2 reports trend-cycle decompositions of the bridged and SIPA series and isolates the cyclical component as the more informative comparison axis. Section OA-E.3 discusses the conceptual and operational reasons why SIPA cannot serve as a level-validation benchmark for the bridge.

OA-E.1 Per-segment series

The headline comparisons in print Section D are at the aggregate level: total registered employment, and total private-plus-public-sector employees. This online appendix decomposes the comparison by employment segment. Total registered employment exhibits a Pearson correlation of 0.86 in levels and 0.539 in first differences across the 36-quarter panel. Restricting to employees (CAT_OCUP = 3 in the EPH; private-plus-public registered wage workers in SIPA) yields level correlation 0.80 and first-difference correlation 0.420.

The independent-worker comparison (employers and self-employed in EPH; monotributistas plus autónomos in SIPA) is more conceptually distant for three reasons. First, SIPA records active monthly contributions, while the EPH unit-level indicators measure status at the survey reference week. Second, monotributistas can be active in SIPA without satisfying the EPH unit-level criteria for any given quarter — a registered monotributista who has not invoiced in the reference week may be classified as informal under the bridge. Third, the EPH covers 31 urban agglomerates while SIPA aggregates national totals, and the rural and small-town independent-worker population is in SIPA but outside the EPH frame. The corresponding correlations are accordingly lower than the employee comparison; the replication package reports them quarter by quarter.

The EPH-to-SIPA ratio of formal worker counts averages approximately 0.59 in levels across the panel and is stable over time. The stability of this ratio suggests that the bridge's level mismatch with SIPA is structural rather than evolving; the structural component is discussed in Section OA-E.3.

OA-E.2 Trend-cycle decomposition

The level correlation of 0.86 reported above is dominated by shared cyclical growth: both series rise with the formal labour force over the panel, so any series that grows roughly in proportion to formal-sector expansion will exhibit a high level correlation with SIPA. The first-difference correlation of 0.539 isolates quarter-to-quarter co-movement net of trend and is the more informative validation statistic for inferential purposes.

A complementary decomposition uses an HP filter (smoothing parameter $\lambda = 1,600$ for quarterly data) to separate trend from cyclical components in log levels. The trend correlation between bridged and SIPA total formality is high; the cyclical correlation, which is the relevant statistic for inferential analyses of changes in formal employment, is moderate but positive. A 4-quarter rolling first-difference correlation tracks the same temporal pattern as the headline first-differences: weakest co-movement during structural breaks (the 2020 pandemic disruption) and strongest during normal-cycle periods. This is the expected pattern for a self-reported survey-based measure compared with administrative records, where the survey lags the administrative source during sharp regime shifts and tracks it more closely during stable periods.

The empirical implication is that researchers using the bridged series for inferential analyses of changes in formality should cite the first-difference correlation (or the cyclical correlation from the HP decomposition) as the relevant validation statistic, not the level correlation. The level correlation is not an informative test of the bridge.

OA-E.3 Limitations of SIPA as a level-validation benchmark

SIPA cannot serve as a level-validation benchmark for the bridge for four reasons.

Geographic coverage mismatch. The EPH covers 31 urban agglomerates representing approximately 60–65% of the national urban population and approximately 50–55% of the total national population. SIPA aggregates the entire national workforce. Even if the EPH bridge were perfectly calibrated, the EPH-to-SIPA ratio of formal worker counts would equal the population coverage ratio in expectation only if formality rates were spatially uniform. Empirically they are not: formality is higher in urban centres covered by the EPH than in non-EPH agglomerates and rural areas, so the EPH-to-SIPA ratio should in principle exceed the population coverage ratio. The observed ratio of approximately 0.59 is below this expectation, suggesting that the bridge classifies fewer non-wage workers as formal than SIPA records as registered.

Conceptual distance for non-wage workers. The bridge's unit-level criteria classify the production unit at the EPH reference week. SIPA records active contributions in the reference month, including from monotributistas in lower categorías (A, B) who may invoice irregularly and from autónomos with administrative dispensations. A self-employed worker registered in SIPA who has not invoiced in the survey week may be classified as informal

under the bridge. This produces a directional level mismatch: SIPA can register a worker the bridge classifies as informal, but the reverse is rare.

Self-reporting versus administrative capture. The EPH formality indicators are self-reported by the worker; SIPA captures employer-side declarations and direct contributions. Self-reporting can introduce bias in either direction: workers in informal arrangements may report registration that does not exist administratively, while workers in marginally formal arrangements (very short-term contracts, household-based monotributistas) may not report registration that does. The expected direction of the net bias on the EPH-to-SIPA ratio is empirically ambiguous.

Compositional heterogeneity within “registered” status. SIPA aggregates several distinct registration categories: private-sector employees with employer contributions, public-sector employees, monotributistas (with internal categories A–K), autónomos, and rural workers under specific regimes. The EPH’s unit-level criteria do not map cleanly onto this heterogeneity: the bridge’s formal/informal binary aggregates a broader population than SIPA’s “registered employment” concept. We treat the change correlations of Section [OA-E.2](#) as the more informative validation axis precisely because they are robust to this compositional mismatch.

OA-F Extended Discussion

This online appendix consolidates extended discussion material referenced in the main text. Section [OA-F.1](#) situates the bridging architecture within a broader literature on survey harmonization, latent-factor models, and bridging by multiple imputation. Section [OA-F.2](#) discusses the methodological choice of household-clustered bootstrap over alternative uncertainty-quantification schemes. Section [OA-F.3](#) provides guidance for applied users of the bridged series.

OA-F.1 Extended literature on survey harmonization and bridging

The bridging literature spans several methodological families that the print introduction (Section ??) summarizes only briefly. This subsection expands those references and clarifies how the architecture in this paper relates to each.

Splicing for instrument refinements. INEGI’s transition from ENOE to ENOE-N (INEGI, [2020](#), [2022](#)), DANE’s GEIH redesigns (DANE, [2023](#)), and the U.S. Census Bureau’s CPS sample-redesign procedure (Bureau of Labor Statistics, [2014](#)) are well-documented examples of splicing under stable conceptual content. Each treats the redesign as a Type 2 change in the typology of print Section 2: the post-redesign instrument refines or modifies measurement of the same target, and standard splicing reproduces the earlier series under the new definition by linking through an overlap window. Eurostat’s documentation for the 2021 LFS regulation (Eurostat, [2016](#)) similarly treats the bulk of the redesign as Type 2, with limited Type 3 elements introduced through the platform-work and atypical-labour modules.

Statistical data fusion and multiple imputation. D’Orazio et al. ([2006](#)) formalize the data-fusion problem in which two surveys with partially overlapping content are combined to recover a joint distribution. The conditional-independence assumption implicit in their framework is the analogue of the structural-stability assumption discussed in print Section 6. Multiple-imputation bridging (Rubin, [1987](#); Si & Reiter, [2013](#); van Buuren, [2018](#); Yang & Reiter, [2024](#)) provides an alternative uncertainty-propagation mechanism through repeated draws of the partially observed variable. Our architecture differs from MI bridging on two design points. First, the predictive stage in Layer 3 is anchored on a rich overlap window (four quarters, $\sim 65,000$ observations) rather than on partially observed cross-sections; this permits explicit calibration evaluation against ground truth and produces a target variable that is reproducible from primitives. Second, the hybrid composition rule of print Equation (1) preserves observed classifications wherever the survey provides them, rather than imputing draws across the full panel; this restricts model dependence to the structurally uncovered universe by construction. The two approaches are complementary in principle: the architecture proposed here is amenable to MI implementation in Layer 3 if a researcher prefers to propagate uncertainty through draws of the formality outcome in the predicted block rather than through the LASSO-coefficient bootstrap of print Section 5.

Latent-factor models and persistent heterogeneity. The latent-factor stage in print Section 4.1 follows the heterofactor approach of Sarzosa and Urzúa ([2016](#)), which identifies cognitive and socioemotional skills from observable proxies under standard non-Gaussian identification (Kotlarski, [1967](#)). Interactive fixed-effects panel-data models provide a related approach to recovering persistent unobserved heterogeneity but require longitudinal panel structure that the cross-sectional EPH does not provide. State-space and dynamic-factor splicing methods address regime breaks in latent processes and are conceptually related to our backcasting stage, but operate on a different identifying restriction (unobserved-component stability across the break rather than observable-predictor stability across the break). Hierarchical bridging frameworks have also been developed in

other survey traditions (Demographic and Health Surveys, household income and consumption surveys); they share the conceptual goal of reconciling instrument generations within a single coherent framework but typically operate on Type 1 and Type 2 changes rather than on Type 3.

The ICLS framework as concept basis. The substantive concept underlying the EPH’s 2024Q4 redesign is the ILO’s ICLS framework for measuring informal employment (ILO, 2013). Parallel implementations of this framework across ILO member countries produce broadly comparable Type 3 redesigns, and the bridging architecture proposed here is portable to those redesigns under the conditions in print Section 6.

OA-F.2 Methodological note on uncertainty quantification

The print Section 5 bootstrap propagates within-overlap LASSO sampling uncertainty to the historical series at fixed penalty λ_{1se} and fixed calibration threshold τ_{cal} . This subsection documents the design rationale for that choice and the alternatives considered.

Why household-clustered. The EPH samples households (not individuals) as primary sampling units, with multiple individuals per household interviewed in the same wave. Within-household correlation in the formality outcome and in the relevant covariates is non-negligible. Bootstrap procedures that resample individuals independently of household structure understate sampling uncertainty by approximately $\sqrt{1 + (m - 1)\rho}$, where m is mean household size and ρ is the intra-household correlation. Resampling at the household level (codusu cluster) preserves within-household correlation by construction.

Why pooled rather than block-bootstrapped. An alternative would block-bootstrap by quarter or by half-year, treating temporal clusters as the resampling unit. The temporal-neutrality diagnostics reported in print Section 4.6 confirm that period-specific effects in the predictor space are negligible within the four-quarter overlap; pooling therefore does not violate exchangeability. We adopt the pooled scheme on efficiency grounds: with only four overlap quarters, block-bootstrap variance is dominated by the small number of resamplable blocks rather than by within-block variation across observations, which understates effective resampling resolution.

Alternatives considered. Bayesian bootstrap, parametric bootstrap on the LASSO coefficients, and non-parametric bootstrap with bias correction were considered. The household-clustered non-parametric bootstrap is preferred for its compatibility with standard EPH analytical conventions and for its directness of implementation; for the present application the alternatives produce 95% bands within a small tolerance of one another at every quarter.

What the bands do not propagate. The bootstrap propagates LASSO sampling uncertainty within the overlap. It does not propagate uncertainty about the LASSO penalty λ , the calibration threshold τ_{cal} , the heterofactor specification choice (Model A vs. Model B), or—the binding restriction—the structural-stability assumption that the predictive relationship estimated in the overlap held in pre-2024Q4 quarters. The bands of ~ 3.7 pp on average reported in print Section 5 are therefore a lower bound on total prediction uncertainty. Multiple imputation, in this respect, has the conceptual advantage of integrating over a model-implied posterior, but the practical disadvantage of requiring a model for the structural-stability assumption that no available data can identify; the choice between bootstrap and MI is fundamentally about whether to be transparent about the limits of within-overlap uncertainty propagation (bootstrap) or to provide a smooth integrated band that absorbs assumptions the data cannot confirm (MI).

OA-F.3 Implications for applied users of the bridged series

The bridged series is constructed for use by applied researchers studying labour formality, informality, and registered employment in Argentina across the 2024Q4 redesign boundary. Three points of guidance follow from the construction logic.

Aggregation domain. The bridged series is anchored on the occupied universe (ESTADO==1). Aggregations to the economically active population or the broader 18–60 universe rely on extensions documented in online Section OA-D.1, where the share of inadmissible predictions grows substantially. Researchers using the bridged series should report results conditional on the occupied universe except where the counterfactual interpretation of the predicted-formality margin among the unemployed and inactive is the explicit object of analysis.

Decomposing change across the redesign boundary. For applied analyses of change in formality across 2024Q4—regressions of formality on policy treatments, time fixed effects, panel structure, or difference-in-differences specifications crossing the redesign—the bridged series should be used in place of the legacy proxy. The 8–14 pp legacy-bridge gap is overwhelmingly mechanical (97.9% concept expansion under the decomposition reported in print Section 5), so any analysis using the legacy series for periods crossing the redesign will conflate

concept change with substantive change. The bridged series neutralizes this by applying a consistent definition throughout. The print decomposition provides the basis for users to attribute residual differences to substantive components.

Subgroup analyses. The bridged series carries different uncertainty profiles across occupational segments. Print Table 3 documents that wage employees enter the observed block (zero predictive uncertainty in pre-redesign quarters) while non-wage workers enter the predicted block. Subgroup analyses focused on the wage-employee segment are unaffected by predictive uncertainty in pre-redesign periods; analyses focused on independent workers should propagate the bootstrap bands of online Section OA-D.1 to inferential statistics. The replication package documents this distinction at the observation level through a flag indicating whether the formality classification is observed or predicted.

Comparison to administrative records. Researchers comparing the bridged series to SIPA or other administrative sources should compare changes rather than levels (online Section OA-E). The level mismatch between the bridged EPH and SIPA reflects geographic coverage, conceptual differences in the registration concept, and self-reporting versus administrative capture; it is not an artefact of the bridge that can be corrected by re-tuning the threshold or the model. The change correlations are the relevant validation statistic.